

Shailender Joseph

Naveen Jindal School of Management, The University of Texas at Dallas

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[Personal Website](#) | [LinkedIn Profile](#) | [Google Scholar](#) | [GitHub](#)

EDUCATION

The University of Texas at Dallas, Richardson, TX, USA, 2021–Present

- Ph.D. in Information Systems, 2027 (expected)
- **Academic Advisors:** Dr. Vijay Mookerjee, Dr. Ganesh Janakiraman, and the late Dr. Milind Dawande

Chennai Mathematical Institute, Chennai, India, 2019–2021

- M.Sc. in Data Science, CGPA: 9.28/10

Birla Institute of Technology and Science, Pilani, India, 2005–2018

- Ph.D. in Drug Delivery and Pharmacokinetics, 2011–2018
- M.Pharm (with specialization in Pharmaceutics), CGPA: 8.03/10, 2009–2011
- B.Pharm (Hons.), CGPA: 7.78/10, 2005–2009

PROFESSIONAL EXPERIENCE

- Professional Development Program (PDP) Scientist, Pfizer Healthcare India Private Limited, Chennai, India, 2017–2019

HONORS AND AWARDS

- Selected Participant, ICIS Doctoral Consortium, 2026
- Selected Participant, INFORMS ISS Doctoral Consortium, 2026
- Graduate Research Fellowship, The University of Texas at Dallas, 2021–2022
- Graduate Studies Scholarship, The University of Texas at Dallas, 2021–Present
- Best Research Scholar Award, Sponsored by Bharat Biotech, 2016–2017
- Best Poster Presentation, PEAR–2017, Jawaharlal Nehru Technological University (JNTU) Hyderabad, 2017
- Senior Research Fellowship (SRF), Council of Scientific and Industrial Research (CSIR), 2013–2017

RESEARCH INTERESTS

Topics: Real-time bidding, platform design, cyber-insurance, and AI-enabled decision systems

Methods: Game theory, optimization, control theory, machine learning, and empirical analysis

RESEARCH PAPERS

Under Review

- **Shailender Joseph**, Milind Dawande, Ganesh Janakiraman, and Vijay Mookerjee. “Win-Curves in Real-Time Bidding: A Micro-Foundation” ([SSRN](#)). **Major Revision** at *Decision Sciences*

Working Paper

- **Shailender Joseph**, Ganesh Janakiraman, and Vijay Mookerjee. “Building Demand While Selling Inventory: Dynamic Bidder Management in Digital Out-of-Home RTB Platforms.” *Manuscript in preparation*

Work in Progress

- **Shailender Joseph**, Kai Sun, Ganesh Janakiraman, and Vijay Mookerjee. “Reimagining Digital Markets: Evaluating Platform Dominance and Network Alternatives.” Presented at **CIST 2025**
- **Shailender Joseph**, Ganesh Janakiraman, and Vijay Mookerjee. “Making Cyber-insurance Accessible: Designing Premiums Under Strategic Claiming.” *Analysis underway*

- **Shailender Joseph**, Aditya Karanam, Ganesh Janakiraman, and Vijay Mookerjee. “Retrieve or Reason? Cost-Effective Routing in Hybrid AI Chatbots.” *Dataset obtained from industry partner*

CONFERENCE PRESENTATIONS

“Win-Curves in Real-Time Bidding: A Micro-Foundation”

- CIST 2024, Seattle, WA
- INFORMS 2024, Seattle, WA
- Big XII+ MIS Research Symposium 2025, Ames, IA
- POMS 2025, Atlanta, GA

“Reimagining Digital Markets: Evaluating Platform Dominance and Network Alternatives”

- CIST 2025, Atlanta, GA
- INFORMS 2025, Atlanta, GA
- DSI 2025, Orlando, FL
- POMS 2026, Reno, NV

TEACHING EXPERIENCE

Instructor

- **ITSS 4381 Object-Oriented Programming with Python** – Fall 2024, 2025, and Summer 2026
- Student evaluation: **4.75/5.00** (Fall 2024)

Teaching Assistant (Selected Courses)

- BUAN 6342 Applied Natural Language Processing (Prof. Harpreet Singh) – Spring 2023, 2025
- BUAN 6V99 Special Topics in Business Analytics (Prof. Harpreet Singh) – Spring 2025
- BUAN 6382 Applied Deep Learning (Prof. Harpreet Singh) – Fall 2022
- MIS 6313 Managing IT in the Analytics Age (Prof. Vijay Mookerjee) – Fall 2022
- MIS 6308 Systems Analysis and Project Management (Prof. Srinivasan Raghunathan) – Fall 2021
- BUAN 6341 Applied Machine Learning (Muhammad Sabir) – Summer 2022

SERVICE

Ad-Hoc Reviewer: Journal

- Information Systems Research – 2026
- Information Technology and Management – 2025, 2026
- Dynamic Games and Applications – 2024
- Production and Operations Management – 2023

Ad-Hoc Reviewer: Conference/Workshop

- Conference on Information Systems – 2026
- International Conference on Information Systems – 2026, 2023
- Workshop on Information Technologies and Systems – 2024

Conference Organization

- BizAI Conference, UT Dallas, 2025, 2026 — registration, poster and presentation sessions

PH.D. COURSEWORK

- **IS Seminar Courses:** Game-theoretical Models (Prof. Srinivasan Raghunathan and Prof. Huseyin Cavusoglu), Topics in Industrial Organization, Empirical Methods, and Algorithmic Bias (Prof. Jianqing Chen and Prof. Amit Mehra), Empirical Methods and Causal Inference (Prof. Eric Zheng), Stochastic Control Theory (Prof. Vijay Mookerjee), IS Design Science Models and Knowledge Management (Prof. Sumit Sarkar), Linear and Integer Programming (Prof. Syam Menon)

- **Methodological Courses:** Advanced Managerial Economics, Game Theory, Advanced Statistics and Probability, Optimization, Probability and Stochastic Process, Deterministic Models in Operations Research, Information Economics and Mechanism Design

SELECTED MACHINE LEARNING PROJECTS — CHENNAI MATHEMATICAL INSTITUTE

- **Computer Vision:** Built object-detection, image-localization, and CNN models for chest X-rays, sign-language digits, and emotion recognition, with Bayesian hyperparameter optimization.
- **Natural Language Processing:** Developed language models, knowledge graphs, and a character-level RNN using the CORPUS-19 corpus.
- **Reinforcement Learning:** Designed a TD-SARSA agent for optimal pathfinding.
- **Predictive Modeling and Data Mining:** Built and evaluated classification models using 10-fold cross-validation and implemented Apriori frequent-itemset mining.

PUBLICATIONS FROM FIRST PH.D.

- **Shailender Joseph**, Punna Rao Ravi, Mrinalini Reddy Sirukuri, Avantika Dalvi, and Odapalli Keerthi Priya. [Drug Development and Industrial Pharmacy](#), 2018; 44:1109–19.
- **Shailender Joseph**, Punna Rao Ravi, Paramita Saha, Avantika Dalvi, and Srividya Myneni. [Colloids and surfaces B: Biointerfaces](#), 2017; 158:610–9.
- **Shailender Joseph**, Punna Rao Ravi, Paramita Saha, and Srividya Myneni. [Xenobiotica](#), 2017; 47:1104–11.
- Punna Rao Ravi, **Shailender Joseph**, Uday Sai Ranjan Avula, and Shruthi Anthireddy. [Acta Chromatographica](#), 2015; 27:597–612.
- Punna Rao Ravi, Rahul Vats, **Shailender Joseph**, and Nitin Gadekar. [Acta Chromatographica](#), 2015; 27:281–94.

SKILLS

- **Programming Skills:** Python, R, SQL, MATLAB
- **Other:** Excel, Mathematica, LaTeX

REFERENCES

Vijay Mookerjee

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